A transitive four-point residual fixed set

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle \leq A_n$ be a transitive string C-group whose noncommuting diagram is connected and whose every maximal omission is intransitive. Suppose G_0 has orbits $\{a\}$ and $B = \Omega \setminus \{a\}$, and write

$$\rho_0 = (a\ b)\delta, \qquad X = \Omega \setminus \{a, b\}, \qquad K = \langle \rho_2, \ldots, \rho_{r-1} \rangle.$$

Thus K fixes a, b and centralizes the odd involution δ on X. Let $\mathcal{T}$ be its set of t two-point orbits, let P be their union, and let $F = \mathrm{Fix}_X(\delta)$. Assume

$$n = 2r \geq 14, \qquad |F| = 4.$$

Consequently $n = 2t+6$, $t \geq 5$ is odd, and K has $t+1$ distinguished generators. Put $\sigma = \rho_1$ and $R = \langle \rho_3, \ldots, \rho_{r-1} \rangle$. The permutation σ fixes a and centralizes R, and $G_0 = \langle \sigma, K \rangle$ is transitive on B.

We use the accepted actual block-orbit deficit proof: deleting any distinguished K image on the δ-blocks strictly refines the orbit partition. In particular all those images are independent. We also retain Whiston's independent-length bound $m-1$ in S_m and its equality clause, which identifies the generated group as S_m. The accepted maximal involutory string theorem says that $m-1$ independent string involutions on an odd $m \geq 5$ form the ordinary Coxeter path; its degree-five finite certificate remains a paid dependency. No quotient intersection property or even-degree maximal-string classification is used below.

Lemma 1 (Transport of a natural orbit)**.** *Suppose $U \subseteq F$ is an R-orbit on which the image is the natural S_s, $s \geq 3$. If $\sigma(U) = O \subseteq P$, then O and O^δ are disjoint R-orbits. Their union consists of exactly s pairs in $\mathcal{T}$. Every member of R fixing U pointwise fixes both orbits pointwise, and therefore fixes those s pair-blocks.*

Proof. Both σ and δ commute with R, so they transport its orbits equivariantly. If $O^\delta = O$, the restriction of δ to O centralizes the natural S_s action. This centralizer is trivial: an element fixing one point fixes every point, and the point stabilizer S_{s-1} fixes only that point when $s \geq 3$. Thus δ would fix O pointwise, contrary to $O \subseteq P$. The two orbits are disjoint, and δ pairs them. Equivariance also gives the final assertion. □

Theorem 2. *Under the stated hypotheses, K is not transitive on F.*

Proof. Suppose it is transitive. We distinguish the possible restrictions of the distinguished involutions to the four points of F.

A double transposition occurs on F. There are two distinguished labels whose restrictions generate a transitive group on F. Indeed, two distinct double transpositions already do so. If only one double transposition occurs, transitivity requires a transposition joining its two pairs, and these two suffice. Call these labels the extras and let J be the other $t-1$ labels.

For each label in J, the full omission still contains the two extras and is transitive on F. Therefore its pair-block orbit partition must change. Its pair image cannot belong to the subgroup on the other pair images. In particular the $t-1$ pair images in J are independent. Whiston makes their group S_t, and the accepted odd-degree extremal string theorem makes

them an ordinary path. Consecutive path marks do not commute, so their indices form an interval in the full string. Write its pair vertices in path order as $C_1, \dots, C_t$.

Each extra has trivial pair image. To see this, omit any path mark. The remaining path has precisely the two orbits separated by that cut. If either extra joined these two orbits, the full omission would again be transitive on the pair-blocks and on F, contradicting the actual block-orbit deficit. Thus each extra preserves every path cut. Their common setwise stabilizer fixes every C_i.

If ρ_2 is an extra, R contains all of J. Its pair image is transitive, so its point orbits in P have size t or $2t$: each such orbit meets every two-point block in the same positive number of points. All its other orbits have size at most four. Hence P is precisely the union of its orbits of size at least five, and σ preserves P. So does K, contradicting transitivity of G_0 on B.

Otherwise J starts at ρ_2, and the two extras η, θ occur consecutively at its far end, in that order. On F they cannot both be transpositions, as two transpositions do not act transitively on four points. They cannot both be double transpositions either: these commute on F, and their trivial pair images make their restrictions to P products of pair flips, which commute too. They would commute globally, disconnecting the full string. Exactly one extra is therefore a transposition on F.

The outer extra θ commutes with every path mark. Its flip vector is constant on the t pairs. If it is odd on F, its evenness globally forces it to flip every pair, since t is odd. If it is even on F, the inner extra η is odd there. This inner extra commutes with all but the last path mark, so its flip vector is constant on $C_1, \dots, C_{t-1}$. Since $t-1$ is even, its global evenness forces a flip on C_t.

The group R now has pair-block orbits $\{C_1\}$ and $\{C_2, \dots, C_t\}$. It is transitive on the points over the latter orbit: in the first case θ flips both points in each block; in the second η flips C_t, and pair transitivity spreads this within-block connection to every block. The resulting point orbit has size $2(t-1) \geq 8$. The extras make F an R-orbit of size four. All other orbits have size at most two, so F is its unique four-point orbit. Centralization forces σ to preserve F, again impossible. This argument includes $t = 5$; a possible four-point row is eliminated by the explicitly forced flip, not by an even-degree rigidity claim.

All nonidentity restrictions to F are transpositions. Their edge graph is connected. No vertex has three incident distinct edges: the three corresponding transpositions would pairwise fail to commute, whereas three distinct positions in a string cannot be pairwise adjacent. The graph is therefore a path or a four-cycle. A four-cycle is impossible, since its four noncommuting adjacency requirements would embed a cycle in the path of generator positions. Thus the graph is a three-edge path.

There is exactly one label on each edge. Choose a label on the middle edge. Labels on both outside edges must occupy its two neighboring positions. A second middle-edge label would have to neighbor both those positions, and a second outside-edge label would have to occupy one of the same already occupied neighboring positions. Neither is possible. We therefore have exactly three consecutive labels η, ζ, θ acting as the ordinary S_4 path on F; all the remaining $t-2$ labels, denoted M, fix F pointwise.

The pair images of M are independent. If one belonged to the subgroup on the other pair images, its full block image would belong to that subgroup too, because every M label fixes F. This contradicts the accepted block-image independence. Every subfamily remains independent.

First suppose ρ_2 belongs to M. Then R is transitive on F with natural image S_4. If σ preserved F, so would G_0. It must therefore send F to a different R-orbit in P; neither a nor b can belong to an orbit of size four. Lemma 1 gives four pair-blocks fixed by all the $t-3$ labels in $M \setminus \{\rho_2\}$. Their independent pair images consequently act on just $t-4$ points. Whiston bounds their number by $t-5$, a contradiction. This also covers $t = 5$, when that action is trivial.

It follows that the three F labels are ρ_2, ρ_3, ρ_4. The group R has a natural three-point orbit $U \subset F$ and fixes the fourth point z. If $\sigma(U)$ were different from U, it would lie in P. Lemma 1

would give three pair-blocks fixed by all $t-2$ labels of M, contradicting the bound $t-4$ on the remaining $t-3$ pair-blocks. Thus σ preserves U.

The point $\sigma(z)$ cannot be in U, by bijectivity, or equal a, since σ fixes a. If it were z, then G_0 would preserve F; if it were b, the involution σ would interchange z, b and G_0 would preserve $F \cup \{b\}$. Therefore $x = \sigma(z)$ lies in P. Since R fixes z and commutes with σ and δ, it fixes both x and x^δ. Write their pair as C.

All $t-2$ independent pair images of M fix C. Equality in Whiston's bound now makes their image the full S_{t-1} on the other pairs. The generator $\zeta = \rho_3$ commutes with every member of M and fixes C pointwise, since it belongs to R. Its pair image centralizes this S_{t-1} and fixes C, so is trivial. Its flip vector is constant on the other $t-1$ pairs. That number is even; hence ζ is even on P, while its restriction to F is a transposition. It fixes a, b, so is odd globally, the final contradiction. □

Attribution and exact boundary. The singleton-end strategy comes from Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Section 5, especially Proposition 5.29 on physical page 38 and the subsequent intransitive-tail branches. The accepted residual deficit is due to the preceding project continuation; Whiston and the full odd-degree extremal-string proof, including its old degree-five certificate, remain retained inputs. The two-generator fixed-set split, orbit transport, and final parity argument are the present addition.

This note excludes only the stated transitive four-point residual case. It does not assume $G_0 = A_B$, any smaller-degree alternating induction, or faithfulness/IP of a pair quotient. It does not exclude additional head configurations by convention. Together with the separately reviewed small-orbit exclusions it can close $|F| = 4$ in that precise endpoint branch, but larger fixed sets, other source branches, rigidity and table obligations remain separate. No full L044 recovery, original count, compression factor, conjecture answer, priority or new formalization is claimed. There is no new finite premise or numerical search.